

Name: Team Timbuktu

Date: 1/23/25

Element H: Prototype testing and data collection plan

To create a prototype testing and data collection plan for the toy enrichment hanger for elands, the first step is to ensure that the testing plan addresses the high-priority design requirements. These requirements include the ability to hold a 100-pound toy securely and stably above an 8-foot fence while ensuring the structure is safe for both the animals and handlers. Materials and joints will be tested for strength and durability under stress, including exposure to environmental conditions such as wind and rain. Additionally, the enrichment hanger's functionality will be evaluated to ensure ease of installation, maintenance, and the capacity to interact with the toy without compromising stability. These requirements align with the project's goal to provide a safe and enriching experience for the elands while maintaining a robust structure.

The testing plan involves a combination of physical and mathematical modeling to ensure reliability and effectiveness. Physical testing will include load-bearing experiments, where the prototype will support incremental weights up to and beyond 100 pounds to determine failure thresholds. Simulated wind resistance tests will assess stability under different environmental conditions. Mathematical modeling will complement physical testing by simulating stress distribution on the hanger's frame using weights. These tests will provide critical data on the prototype's performance and identify potential points of failure or areas for improvement. Conducting physical tests ensures a comprehensive understanding of the design's capabilities.

The quantity of tests will be defined based on the instructional context, allowing for iterative improvements through multiple rounds of evaluation. The prototype will undergo at least five cycles of load testing and environmental condition simulations. The logical flow

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of the testing plan ensures that initial tests focus on basic structural integrity before advancing to more complex evaluations like specific joint integrity. Input from field experts, such as zookeepers and structural engineers, will be sought to validate the practicality and safety of the prototype. Their feedback will guide refinements, ensuring the final product meets the needs of both the elands and the caretakers while adhering to safety standards

